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From Immediate Answers to Structured Inquiry: A Socratic Method-Enhanced ChatGPT System for Blended Learning Discussions

Hsin-Yu Lee[1], Pei-Hua Chen[2], Yueh-Min Huang[2], and Ting-Ting Wu[1]

Abstract

Artificial Intelligence integration with mobile instant messaging (MIM) offers promising support for blended learning discussions, yet current implementations often favor immediacy over pedagogical structure. Despite separate studies on Socratic methods, ChatGPT applications, and MIM platforms, their systematic integration remains unexplored. This study introduces SCMIM (Socratic-based ChatGPT Mobile Instant Messaging) and compares its impact with standard ChatGPT-MIM and traditional MIM through a 12-week randomized trial with 72 master's students. Using mixed-methods analysis of engagement questionnaires, discussion logs, and interviews, we examined behavioral, cognitive, and emotional engagement alongside technology acceptance perceptions. Results showed ChatGPT-MIM enhanced behavioral participation and emotional comfort through immediate support, while SCMIM more effectively fostered cognitive engagement, particularly analytical thinking and knowledge construction. Despite initial adaptation challenges, SCMIM users developed appreciation for systematic questioning, reporting high perceived usefulness and comparable intention to use. These findings demonstrate that effective AI integration requires balancing

[1]Graduate School of Technological and Vocational Education, National Yunlin University of Science and Technology, Yunlin, Taiwan, ROC
[2]Department of Engineering Science, National Cheng Kung University, Tainan, Taiwan, ROC

Corresponding Author:
Yueh-Min Huang, Department of Engineering Science, National Cheng-Kung University, No. 1, University road, Tainan 701 Taiwan, R.O.C.
Email: huang@mail.ncku.edu.tw

immediate support with structured pedagogical guidance to promote both engagement and deeper learning outcomes in educational discussions.

Keywords
blended learning, ChatGPT, mobile instant messaging, socratic method, student engagement, student perceptions

Introduction

The Challenge of Effective Discussion Support in Blended Learning

Blended learning, integrating traditional face-to-face instruction with online components, has become a dominant pedagogical model in higher education (Yen et al., 2018). This approach leverages complementary strengths: face-to-face sessions enable direct interaction and immediate clarification, while online components support extended reflection and flexible engagement. Central to successful implementation are online discussions that bridge physical and digital learning environments, facilitating critical thinking and sustained engagement between sessions (Fisher et al., 2018; Shang, 2021). Research shows students actively participating in online discussions within blended environments significantly outperform those with limited participation (Zimba et al., 2021).

However, current approaches to supporting online discussion face substantial limitations. Traditional mobile instant messaging (MIM) platforms lack mechanisms for immediate instructional support and structured pedagogical guidance (Tang & Hew, 2022). Recent AI integration, particularly ChatGPT, addresses immediate support needs (Huang et al., 2025), yet often favors information delivery over structured cognitive development, potentially promoting passive consumption rather than active knowledge construction (Lee et al., 2024; Wu et al., 2025). This tension between immediate accessibility and learning depth represents a critical challenge for blended environments where online discussions must maintain continuity with face-to-face instruction while promoting independent critical thinking.

The COVID-19 pandemic accelerated blended learning adoption, with most higher education institutions planning expanded hybrid offerings (Mali & Lim, 2021; Tomei et al., 2024), intensifying the need for discussion platforms that balance immediate support with pedagogical structure.

Mobile Instant Messaging: Promise and Challenges

Mobile instant messaging (MIM) applications have emerged as practical tools for supporting online discussion in blended learning, leveraging students' daily familiarity with these platforms (Lee & Wu, 2025; Tang & Hew, 2017). Shang (2021) documented

significant engagement improvements when incorporating MIM into English language instruction within blended frameworks.

However, traditional MIM faces substantial challenges. Li et al. (2023) identified information overload as a critical limitation, where rapid message flows overwhelm students. Additionally, MIM's informal nature often prevents structured argument development and in-depth exploration (Ardèvol-Abreu et al., 2023). Most significantly, traditional MIM lacks mechanisms for immediate instructional support between face-to-face sessions, leading to decreased engagement when students encounter difficulties (Tang & Hew, 2022).

Artificial Intelligence in Educational Discussions

ChatGPT integration into educational platforms offers a promising response to traditional MIM limitations. Recent studies demonstrate various approaches to incorporating AI support into mobile messaging environments (Huang et al., 2025; Lee & Huang, 2024). Basic ChatGPT integration provides immediate responses, addressing information access gaps (Labadze et al., 2023; Lee et al., 2025), with Huang et al. (2025) reporting improved response times and user satisfaction.

Despite these advances, straightforward AI integration exhibits pedagogical limitations. Lee et al. (2024) observed that while AI facilitates immediate information access, it often promotes passive consumption rather than active knowledge construction. Wu et al. (2025) identified student tendency to rely on AI-generated responses without critical analysis. These findings suggest effective educational AI integration requires sophisticated pedagogical frameworks to balance immediate support with intellectual development.

The Necessity of Socratic Method Integration

The limitations in both traditional MIM and basic AI integration point to a fundamental need for structured pedagogical frameworks. The Socratic method, emphasizing systematic questioning, assumption examination, and guided discovery, emerges as particularly suitable (Paul & Elder, 2007).

Modern implementations have evolved beyond simple question-answer sequences to incorporate systematic inquiry with structured support (Elder & Paul, 2010), remaining particularly relevant for AI-enhanced learning where online discussions must sustain face-to-face instruction's intellectual rigor. Oyler and Romanelli (2014) demonstrated Socratic questioning significantly improved critical thinking and knowledge retention, while Patel (2022) showed how Socratic elements facilitated deeper conceptual understanding.

The necessity of integrating Socratic methodology with AI becomes apparent when examining complementary strengths. Wu et al. (2025) developed the PA-GPT system, demonstrating how AI can effectively implement Socratic questioning through peer assessment roles, achieving remarkable improvements in higher-order thinking by creating structured dialogue encouraging deep reflection. Etori and Gini (2024)

confirmed that combining Socratic principles with conversational AI creates more dynamic learning experiences.

Critically, the Socratic method addresses basic AI integration's primary weakness: passive information consumption. By structuring AI responses through Socratic principles, platforms can maintain immediate support benefits while actively engaging students in knowledge construction that bridges and extends face-to-face learning.

To illustrate fundamental differences, consider responses to: *"What is collaborative learning?"*

ChatGPT-MIM would provide a direct response:

> "Collaborative learning is an educational approach where students work together in groups to achieve shared learning goals. Key characteristics include: (1) positive interdependence among group members, (2) individual accountability for contributions, (3) face-to-face interaction, (4) development of social skills, and (5) group processing of effectiveness. This approach has been shown to improve academic achievement, critical thinking, and social skills development."

In contrast, SCMIM would structure its response through Socratic principles:

> "Collaborative learning involves students working together toward shared goals. Let me help you explore this concept more deeply.
>
> Consider your own group work experiences: What made some collaborations more effective than others? This question helps us examine a key assumption, that simply placing students in groups creates effective learning. Research shows that successful collaboration requires specific conditions: clear roles, mutual accountability, and structured interaction patterns.
>
> Now, think about the implications: How might collaborative learning principles apply to online discussions like this one? What differences might exist between face-to-face and digital collaboration? These connections reveal that collaborative learning isn't just a classroom technique, it's a framework for understanding how people construct knowledge together, applicable across various learning contexts."

This comparison reveals the core distinction: while ChatGPT-MIM delivers comprehensive information efficiently, SCMIM transforms queries into opportunities for guided intellectual exploration.

Research Gap and Objectives

Current research reveals three distinct but separate streams. First, extensive literature documents Socratic method implementations in traditional face-to-face contexts (Elder & Paul, 2010; Oyler & Romanelli, 2014), but these studies focus exclusively on synchronous, human-facilitated environments without addressing scalability or asynchronous contexts. Second, emerging research explores ChatGPT applications in education (Huang et al., 2025; Wu et al., 2023), yet these implementations primarily

emphasize immediate information delivery without incorporating structured pedagogical frameworks (Lee et al., 2024). Third, substantial research addresses MIM platforms in educational settings (Shang, 2021; Tang & Hew, 2022), identifying persistent limitations in providing structured instructional guidance.

Critically, while Wu et al. (2025) demonstrated successful Socratic principles integration with ChatGPT in peer assessment, and Etori and Gini (2024) explored Socratic-ChatGPT combinations for emotional support, no empirical investigation has examined systematic integration of Socratic methodology with ChatGPT within MIM platforms specifically for blended learning discussions. This represents a significant gap because blended learning presents unique challenges requiring discussion platforms that bridge physical and digital spaces while maintaining pedagogical continuity. The absence of research examining how Socratic-enhanced AI can support this context leaves practitioners without evidence-based guidance.

This study addresses this gap by investigating how systematic Socratic methodology integration with ChatGPT influences student engagement and perceptions in blended learning discussions. Through a three-group randomized controlled trial comparing SCMIM (Socratic-based ChatGPT MIM), standard ChatGPT-MIM, and traditional MIM, this research examines:

(1) How does SCMIM affect student engagement dimensions compared to standard ChatGPT-MIM and traditional MIM in online discussion component of blended learning?
(2) How do students perceive SCMIM compared to other platforms using the Technology Acceptance Model within blended learning contexts?

Engagement, encompassing behavioral, cognitive, and emotional dimensions (Fredricks et al., 2004), provides comprehensive insight into how different AI integration approaches influence participation patterns and learning depth. Perceptions, measured through the Technology Acceptance Model (Davis, 1989), are crucial for evaluating user acceptance and sustained adoption. Together, these measures create a holistic evaluation framework capturing both learning processes and user experiences.

The three-group comparative design was specifically selected to create comprehensive understanding of AI integration effects. Traditional MIM serves as baseline, representing current standard practice and allowing measurement of fundamental AI integration impact. ChatGPT-MIM represents prevalent basic AI integration approaches, offering immediate support without pedagogical structure, enabling evaluation of pure AI benefits versus structured support. SCMIM, as experimental treatment, incorporates systematic Socratic methodology with AI capabilities. This deliberate selection allows isolating specific effects of pedagogical structure by comparing both unstructured and structured AI approaches against traditional methods.

The findings aim to provide evidence-based guidance for developing AI-enhanced educational technologies that balance immediate support with structured pedagogical guidance, promoting both engagement and deeper learning outcomes in blended learning environments.

System Implementation

ChatGPT-MIM System Implementation

ChatGPT-MIM integrates ChatGPT with mobile instant messaging to provide AI-enhanced support for blended learning discussions. The system utilizes the Line messaging platform, selected for its widespread adoption among university students in East Asian educational contexts (Steinberg, 2020).

The system architecture (Figure 1) comprises three main components: the Line messaging interface for user interaction, the ChatGPT processing module for query analysis and response generation, and the Firebase data management system for conversation logs and data retrieval.

ChatGPT-MIM offers two core functions: @reflect and @synthesize. The @reflect function enables students to receive immediate AI-supported responses to their questions. When triggered, the query flows through the Line Messaging API to the server, where ChatGPT (GPT-4o-mini with temperature 0.7) processes it and returns a response, which is then logged in Firebase.

The @reflect function employs a straightforward prompt template focused on direct information delivery:

"For the student question '[student_input]', please provide:

1. A clear and accurate explanation of the concept
2. Key definitions and essential points"

Figure 2 demonstrates how ChatGPT-MIM processes a student query, showing its direct response approach that prioritizes clarity and immediacy over pedagogical structure.

The @synthesize function provides comprehensive summaries of discussion threads by retrieving recent messages (typically 500) from Firebase and processing them

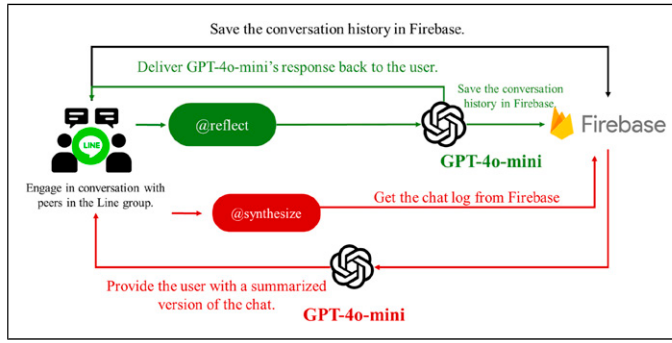


Figure 1. The workflow of ChatGPT-MIM

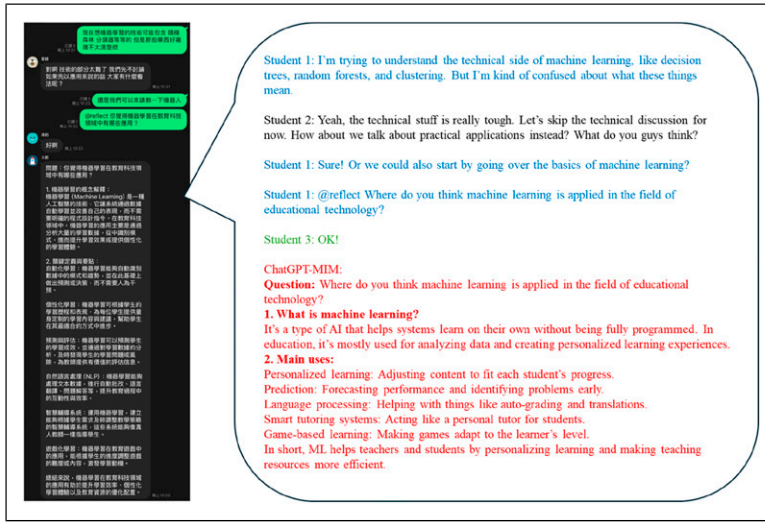


Figure 2. The example of reflecting in ChatGPT-MIM

through ChatGPT (GPT-4o-mini with temperature 0.3). The synthesis prompt maintains the same emphasis on clarity and directness:

"Please analyze the following discussion content and provide:

1. A summary of key points discussed
2. Main conclusions or insights reached
3. Any important questions or issues raised"

Figure 3 illustrates how ChatGPT-MIM synthesizes a discussion thread about collaborative learning technologies, demonstrating the system's ability to extract and organize key information from extended discussions without incorporating structured pedagogical frameworks.

While ChatGPT-MIM effectively addresses immediate support limitations of traditional MIM platforms, its approach primarily focuses on information delivery rather than structured cognitive development.

The Need for Socratic Method Integration in AI-Enhanced Discussions

SCMIM's development emerged from recognizing fundamental limitations in basic AI integration approaches. ChatGPT-MIM, while effective at delivering immediate information, tends to promote passive consumption rather than active knowledge construction.

The Socratic method offers a valuable framework for addressing these limitations for several reasons. First, as demonstrated by Oyler and Romanelli (2014), Socratic

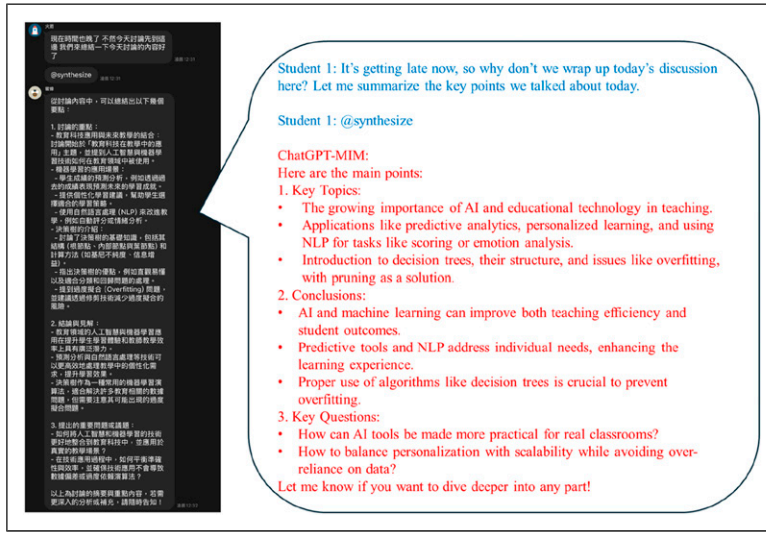


Figure 3. The example of synthesizing in ChatGPT-MIM

questioning specifically develops critical thinking skills by guiding learners through systematic concept examination rather than presenting finished conclusions. Second, the method's emphasis on assumption examination creates natural opportunities for metacognitive development, helping students become aware of their thinking processes (Elder & Paul, 2010). Third, its focus on implication exploration encourages students to connect concepts with broader applications, counteracting the tendency toward isolated information acquisition often seen in basic AI interactions.

The integration of Socratic methodology with AI capabilities creates a particularly powerful combination by addressing the limitations of each approach independently. Traditional Socratic teaching requires continuous instructor presence, making implementation challenging in asynchronous online discussions. AI integration addresses this limitation by providing consistent, scalable Socratic guidance. Conversely, basic AI responses tend toward direct information provision without structured cognitive scaffolding. Socratic principles transform AI responses into guided cognitive experiences that promote active learning.

SCMIM System Implementation and Socratic Method Integration Process

SCMIM builds upon ChatGPT-MIM's technical foundation while introducing sophisticated pedagogical elements through structured prompt engineering. As shown in Figure 4, while the basic system architecture remains similar, SCMIM incorporates a Socratic enhancement layer that transforms simple queries into opportunities for deeper learning and critical thinking development.

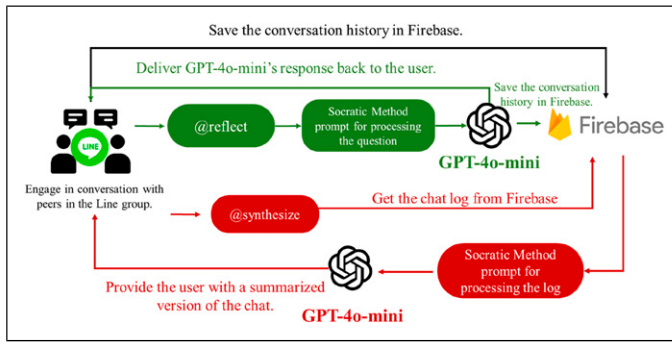


Figure 4. The workflow of SCMIM

The technical implementation maintains the core components of Line Messaging API, ChatGPT, and Firebase. However, SCMIM utilizes GPT-4o-mini with temperature 0.4 for @reflect and GPT-4o-mini with temperature 0.3 for @synthesize, settings determined through extensive testing to balance response accuracy with elaboration needed for effective Socratic dialogue.

The integration of Socratic methodology into SCMIM followed a systematic four-phase development process:

(1) **Principle Analysis and Adaptation:** Identifying core Socratic principles from educational literature (Elder & Paul, 2010; Paul & Elder, 2007) and adapting them to mobile instant messaging constraints. This phase established three foundational principles: systematic questioning, assumption examination, and implication exploration.

(2) **Prompt Engineering:** Translating Socratic principles into specific prompt structures to guide AI response generation. This phase required careful technical implementation that balanced pedagogical sophistication with executable instructions. The translation process involved constructing three distinct layers of prompt architecture. First, the content analysis layer instructs the model to decompose student queries by identifying the main concept, mapping related ideas, and recognizing underlying assumptions before generating any response, ensuring that responses address not just the surface question but the conceptual territory it inhabits. Second, the structured response layer explicitly sequences response components, directing the model to: (a) begin with direct explanation for immediate comprehension, (b) transition to assumption discussion using phrases like "Consider that…"or "A key assumption underlying this concept is…", and (c) conclude with extension and implications using connective language such as "This connects to broader concepts of…" This sequencing mirrors natural Socratic dialogue progression while maintaining coherence within a single response. Third, the reflection prompt layer generates questions tailored to specific content discussed, instructing the model to

formulate 2-3 specific questions that encourage students to examine concept relationships, evaluate identified assumptions, and consider applications in course-relevant contexts, ensuring that reflection prompts emerge naturally from the content rather than appearing as disconnected add-ons. The temperature setting of 0.4 for SCMIM's @reflect function was deliberately chosen through iterative testing: lower temperatures (0.1-0.2) produced overly formulaic responses lacking natural pedagogical flow, while higher temperatures (0.6-0.8) introduced excessive variation that sometimes compromised the systematic structure essential to Socratic methodology. The selected value ensures responses maintain structured pedagogical intent while allowing sufficient flexibility for context-appropriate elaboration.

(3) **Response Structure Optimization:** Creating optimal response structures that balance immediate information needs with deeper learning scaffolding, determining the most effective sequence for guiding students from basic understanding to critical analysis.

(4) **Implementation Refinement:** Iterative testing and refinement to ensure Socratic principles maintained coherence and effectiveness, including adjusting model parameters, refining prompt structures, and validating response quality.

Table 1 illustrates how Socratic principles were systematically implemented in SCMIM's prompt design.

Based on this framework, SCMIM implements a comprehensive prompt structure that ensures each response provides both immediate understanding and scaffolds for further reflection:

"For the student question '[student_input]', please generate a response that includes:

1. Content Analysis:

- Identify the main concept and key aspects
- Map related concepts and applications
- Note underlying assumptions

2. Structured Response:

Table 1. Socratic Principles in SCMIM Prompt Design

Socratic principle	Implementation purpose	Prompt implementation
Systematic questioning	Guide students through logical concept examination	Clear explanation followed by prompts for self-questioning about key concepts
Examining assumptions	Help students identify and evaluate underlying beliefs	Explicit discussion of assumptions and evidence, with reflective prompts
Exploring implications	Encourage broader thinking and application	Connection to related concepts and real-world applications

(a) Direct Answer:
- Clear explanation of the concept
- Definition and key characteristics
- Concrete examples

(b) Assumption Discussion:
- Identify key assumptions in understanding this concept
- Provide evidence supporting or challenging these assumptions
- Highlight common misconceptions

(c) Extension and Implications:
- Connect to broader concepts and applications
- Suggest areas for further exploration
- Present alternative perspectives

3. Reflection Prompts:

- Questions about concept relationships
- Prompts to examine assumptions
- Questions about broader implications"

This structure enables SCMIM to provide comprehensive support within a single response, including clear explanation, discussion of assumptions, connections to broader concepts, and thoughtful prompts for further reflection.

Figure 5 demonstrates this approach, showing how a single SCMIM response provides both immediate understanding and scaffolds for deeper thinking.

Similarly, the @synthesize function adapts to the single-response constraint while maintaining Socratic principles through a specialized prompt structure:

"For the provided discussion thread, generate a response that:

1. Analysis and Summary:
   - Synthesize key concepts and relationships
   - Identify patterns and themes
   - Note important assumptions and evidence
2. Integrated Discussion:
   - Present main conclusions with supporting evidence
   - Highlight different perspectives and approaches
   - Connect ideas across discussion threads
3. Reflection Framework:
   - Provide questions for deeper concept exploration
   - Suggest areas for assumption examination
   - Offer prompts for broader application"

Figure 6 shows how this structure helps students engage more deeply with discussion content through a single, well-structured synthesis.

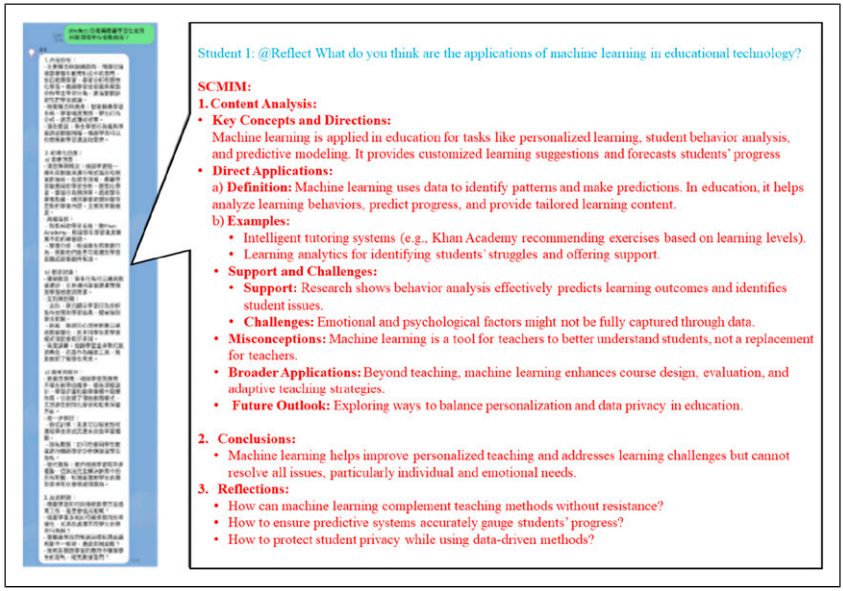


Figure 5. The example of reflecting in SCMIM

The comparison between ChatGPT-MIM and SCMIM responses demonstrates how the structured integration of Socratic principles transforms simple answers into comprehensive learning scaffolds. This systematic integration represents a fundamental shift in how AI can support educational discussions, transforming AI from a simple information source into a cognitive scaffolding tool that promotes deeper engagement with course content while developing advanced learning capabilities.

Methodology

Participants

This study employed a randomized controlled trial design at a university in southern Taiwan. Based on power analysis using G*Power 3.1 (Faul et al., 2007) (one-way ANOVA, $f = 0.40$, $\alpha = .05$, power = .80), 72 master's students (39 females, 33 males; aged 22-28 years, $M_{age} = 24.5$, $SD = 1.3$) enrolled in an Advanced Educational Technology Applications course were randomly assigned to three groups: Treatment Group 1 (TG1, $n = 24$) using SCMIM, Treatment Group 2 (TG2, $n = 24$) using ChatGPT-MIM, and Control Group (CG, $n = 24$) using traditional MIM.

Baseline data verified randomization effectiveness, with no significant differences between groups in GPA ($F(2,69) = 0.38$, $p = .68$), technology proficiency ($F(2,69) = 0.42$, $p = .66$), gender distribution ($\chi^2(2) = 0.32$, $p = .85$), or prior online learning experience ($\chi^2(4) = 2.87$, $p = .58$). All groups received identical course materials and

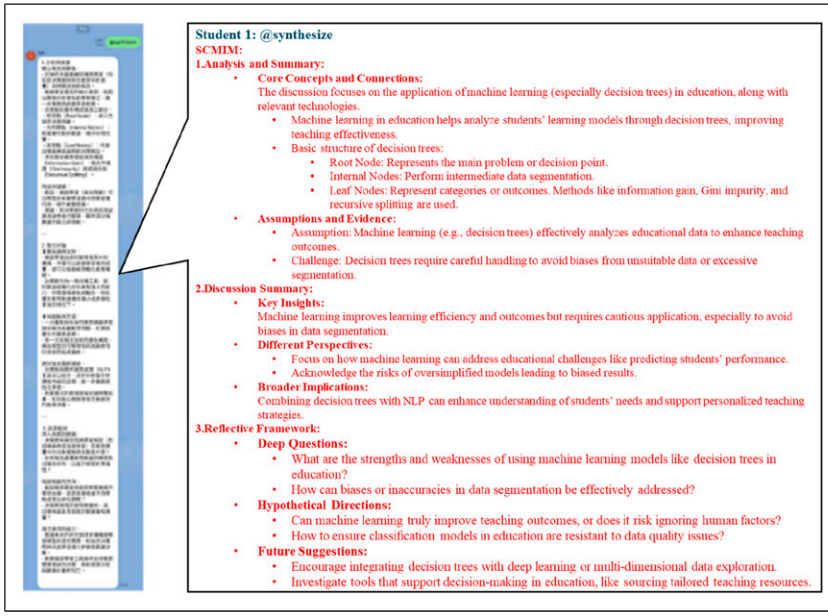


Figure 6. The example of synthesizing in SCMIM

instruction, with the only variation being the messaging tool used for online discussions. The study received approval from the university's Research Ethics Committee, with participants providing informed consent.

Experimental Procedure

The experiment was conducted within a 12-week course comprising three modules: emerging technology integration, instructional design innovation, and learning analytics applications. As illustrated in Figure 7, the procedure consisted of three phases: preparation, implementation, and assessment.

The preparation phase (Weeks 1-3) established baseline conditions and ensured participants' readiness. Baseline data verified randomization effectiveness, and participants completed the engagement questionnaire to establish pre-intervention levels. All participants received comprehensive training on their assigned discussion tools. SCMIM (TG1) and ChatGPT-MIM (TG2) systems provided identical commands (@reflect and @synthesize) but differed in response structures: SCMIM incorporated Socratic principles (systematic questioning, assumption examination, and implication exploration), while ChatGPT-MIM provided direct, non-structured responses.

The implementation phase (Weeks 4-10) followed a structured weekly blended learning cycle. Students engaged with course materials through the Learning Management System, then participated in online discussions with at least two substantive posts and two peer responses. TG1 and TG2 participants utilized their respective

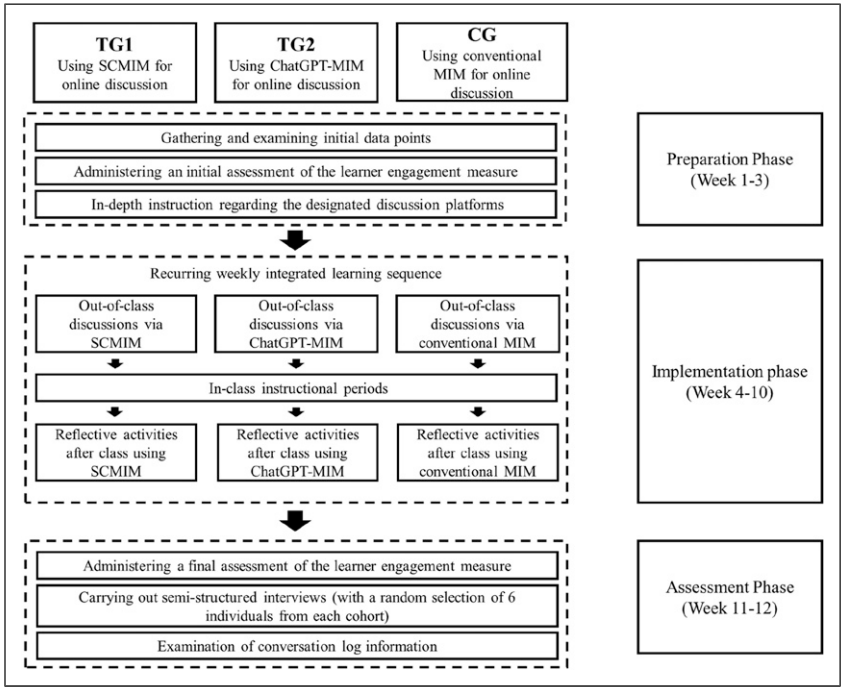


Figure 7. Experimental procedure

AI-enhanced tools for concept clarification (@reflect) and discussion summary (@synthesize), while CG participants engaged in traditional text-based discussions. Chat histories were systematically archived throughout.

Weekly classroom sessions (2.5 hours each) provided face-to-face learning through interactive lectures, hands-on activities, and reflection. Following each session, students completed reflection assignments within 48 hours and continued online discussions.

The assessment phase (Weeks 11-12) concluded the intervention with comprehensive measures including post-intervention engagement questionnaires and semi-structured interviews with randomly selected participants ($n = 6$ per group). Throughout the experimental period, systematic documentation maintained consistent support across all groups.

Data Collection and Analysis

Engagement Measurement. The study adopted a mixed-methods approach to evaluate engagement across all three conditions, examining behavioral, cognitive, and emotional dimensions as illustrated in Figure 8.

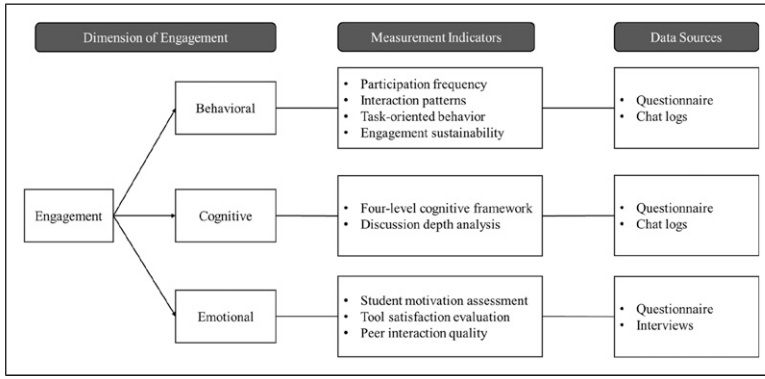


Figure 8. Engagement measurement framework

Learner engagement was measured using Sun and Rueda (2012)'s validated 15-item questionnaire adapted for technology-enhanced learning environments. This 5-point Likert scale instrument assessed behavioral engagement (3 items), emotional engagement (5 items), and cognitive engagement (7 items), demonstrating strong internal consistency (Cronbach's alpha: 0.87, 0.85, and 0.89 respectively).

Behavioral engagement was further analyzed through interaction logs following Tang and Hew (2022)'s methodology, examining participation frequency, interaction patterns, task-oriented behavior, and engagement sustainability.

Cognitive engagement was evaluated through the questionnaire and content analysis of discussion logs. Following Huang et al. (2025), interactions were categorized into four cognitive levels: operation, wayfinding, sense-making, and innovation. Table 2 presents the analytical framework used for discussion content analysis, which employed a deductive coding approach based on established cognitive engagement taxonomy (Huang et al., 2025). This framework was selected because it provides explicit operational definitions aligned with our research focus on cognitive processing levels in AI-enhanced discussions.

Two researchers independently coded 25% of the discussion content using this predetermined framework, achieving a Cohen's kappa coefficient of 0.85. Disagreements were resolved through discussion until consensus was reached, with specific attention to boundary cases between sense-making and innovation levels. Following successful reliability establishment, one researcher coded the remaining data, with the second researcher reviewing 10% of subsequently coded data to ensure coding consistency throughout the analysis process.

Emotional engagement was assessed through semi-structured interviews with randomly selected participants (n = 6 per group). These 30-min interviews explored participants' experiences with their assigned discussion tools through specific questions:

(1) "How does the discussion tool influence your learning motivation?"

Table 2. Analytical Framework for Discussion Content

Code	Definition	Example
Operation		
Sharing information	Presenting factual content or resources	*"I've attached an article on AI strategies."*
Seeking information	Posing questions for facts or explanations	*"What distinguishes synchronous from asynchronous tools?"*
Wayfinding		
Basic commenting	Brief responses without analysis	*"Your assessment perspective is interesting."*
Elaboration request	Requesting additional context	*"Could you elaborate on learning analytics here?"*
Extended response	Comprehensive answers with rationale	*"AR requires device accessibility and curriculum alignment."*
Information synthesis	Organizing information into summaries	*"Four digital literacy approaches emerged."*
Sense-making		
Comparative analysis	Contrasting elements to identify patterns	*"AI evaluations provide immediate feedback unlike traditional methods."*
Critical evaluation	Assessing validity using criteria	*"The framework is theoretically strong but faces practical constraints."*
Innovation		
Reflective integration	Connecting concepts with experiences	*"These techniques unexpectedly enhanced student motivation."*
Novel application	Proposing original solutions	*"We could develop hybrid assessment combining AI and peer review."*

(2) "What aspects of the system's responses enhance or hinder your engagement?"
(3) "How does the tool affect your interaction with peers?"
(4) "Describe your experience with the tool's questioning or response style."

Interview data underwent thematic analysis using MAXQDA software, employing an inductive approach to identify emergent themes within the overarching framework of emotional engagement. Two researchers independently coded 25% of transcripts (Cohen's kappa = 0.83), with disagreements resolved through collaborative discussion and reference to original interview contexts. The final coding scheme emerged through iterative refinement, with three primary themes identified: Learning Support Experience, Discussion Quality, and Learning Confidence.

Data analysis employed separate ANCOVAs for each engagement dimension (behavioral, cognitive, emotional) with pre-test scores as covariates. Following Fredricks et al. (2004)'s theoretical framework and established engagement research conventions (Sun & Rueda, 2012; Tang & Hew, 2022), separate analyses were preferred over MANCOVA because these dimensions represent distinct theoretical

constructs rather than indicators of a single latent variable. Our research questions explicitly predict differential effects across dimensions, ChatGPT-MIM enhancing behavioral/emotional engagement through immediate support, while SCMIM promoting cognitive engagement through structured questioning. Separate analyses allow direct testing of these dimension-specific hypotheses and clearer interpretation of theoretically meaningful effects (Huberty & Morris, 1992). For each ANCOVA, pairwise comparisons used Bonferroni adjustment. The partial η^2 values reported throughout the Results section represent effect sizes, with values of .01, .06, and .14 typically interpreted as small, medium, and large effects respectively (Cohen, 2013). Our observed effect sizes (ranging from .14 to .30) indicate medium to large practical significance, suggesting that the differences between platforms represent meaningful impacts on student engagement beyond statistical significance. Additional analyses included chi-square tests for interview themes, one-way ANOVA with Tukey's HSD for behavioral indicators, and thematic analysis for emotional engagement patterns.

Learners' Perceptions. Student perceptions were measured using semi-structured interviews based on Davis (1989)'s Technology Acceptance Model (TAM). The interview protocol was validated by three experts in educational technology and AI-enhanced learning. The analysis employed a hybrid deductive-inductive approach: the four TAM dimensions (Perceived Usefulness, Perceived Ease of Use, Attitude Towards Use, and Behavioral Intention to Use) provided the deductive framework, while specific themes within each dimension emerged inductively from participant responses. The interview protocol examined four key dimensions:

(1) Perceived Usefulness (PU)
   - "How did the discussion tool enhance your learning in blended environments?"
   - "Which features contributed most significantly to your learning process?"
(2) Perceived Ease of Use (PEOU)
   - "How did the discussion tool enhance your learning in blended environments?"
   - "Which features contributed most significantly to your learning process?"
(3) Attitude Towards Use (ATU)
   - "What difficulties did you encounter while using the discussion system?"
   - "How would you describe the learning curve of the tool's features?"
(4) Behavioral Intention to Use (BIU)
   - "Would you choose to use this tool in future blended learning courses?"
   - "What learning contexts would be most suitable for this tool?"

All interviews were audio-recorded, transcribed verbatim, and analyzed using MAXQDA software following Braun and Clarke (2006)'s thematic analysis framework. Two researchers independently coded 25% of the transcripts, achieving an inter-

rater reliability coefficient of 0.84. Disagreements were resolved through iterative discussion, with researchers revisiting original transcripts to ground interpretations in participant language. Once consensus was achieved on the initial 25%, one researcher coded the remaining data, with periodic verification meetings to ensure coding consistency and discuss emerging patterns.

Chi-square analyses examined the distribution of perception themes across platforms, with Bonferroni-adjusted z-tests for post-hoc comparisons, enabling systematic assessment of how different AI integration approaches influenced student perceptions in blended learning environments.

Ethical Considerations and AI Risk Assessment

Following the RAISE (Reporting AI Studies in Education) framework (Allison, 2025), we systematically identified and addressed potential ethical and practical risks associated with AI integration in this study. The RAISE framework provides structured guidance for transparent reporting of AI studies in educational contexts, ensuring comprehensive consideration of pedagogical, equity, data governance, and technical concerns.

We completed the RAISE Ethics and Risk Matrix (see Appendix A) to document anticipated and observed impacts across four risk themes: (1) Pedagogical Integrity and Learning Impact, (2) Equity, Inclusion, and Accessibility, (3) Data Governance and Learner Rights, and (4) Technical Dependability and System Reliability. Key ethical considerations and mitigation strategies included:

Pedagogical risks were addressed through careful system design. While SCMIM's structured questioning initially created adaptation challenges (58% reported difficulty in early stages), we provided comprehensive three-week training and progressive scaffolding. To prevent AI overreliance observed in ChatGPT-MIM, SCMIM employed Socratic questioning rather than direct answers, encouraging critical thinking. We monitored AI-generated content quality through regular instructor audits and calibrated temperature settings (0.3-0.4) to minimize hallucination risks.

Equity and inclusion considerations guided participant selection and support. Random assignment with baseline verification ensured balanced groups across gender, GPA, and technology proficiency. All participants had access to necessary devices through the widely-adopted Line platform, with equal technical support provided across groups.

Data governance protocols received approval from the university's Research Ethics Committee. All participants provided informed consent after comprehensive explanation of AI's role, data usage, and their rights. Discussion data were anonymized and securely stored through Firebase, with clear documentation of ownership and usage rights limited to research purposes.

Technical reliability was ensured through use of stable commercial platforms (Line, OpenAI, Firebase) with continuous monitoring and technical support. We documented system design and prompt engineering methods comprehensively to facilitate replication and maintained instructor oversight authority for AI-generated content.

This systematic risk assessment and mitigation approach aligns with the RAISE framework's emphasis on transparent reporting of AI studies, ensuring our research maintained ethical standards while investigating pedagogical innovations in AI-enhanced education.

Results

The Impact on Engagement

Analysis across SCMIM, ChatGPT-MIM, and traditional MIM revealed distinct patterns reflecting each system's relative strengths and limitations across behavioral, cognitive, and emotional dimensions of engagement.

Behavioral Engagement. ANCOVA assumptions were verified (Levene's test: $F(2,69) = 0.84$, $p = .436$; homogeneity of regression slopes: $F(2,66) = 0.42$, $p = .659$). ANCOVA revealed significant differences in behavioral engagement ($F(2,68) = 10.63$, $p < .001$, partial $\eta^2 = .24$), as shown in Table 3.

Pairwise comparisons using Bonferroni adjustment revealed ChatGPT-MIM significantly outperformed SCMIM (mean difference = 0.22, $p = .032$), and both AI-enhanced platforms significantly outperformed traditional MIM (ChatGPT-MIM: mean difference = 0.67, $p < .001$; SCMIM: mean difference = 0.45, $p < .001$). Discussion log analysis revealed further differentiation patterns (Table 4).

ChatGPT-MIM generated highest message volume ($M = 1197$), significantly exceeding SCMIM ($M = 952$) and traditional MIM ($M = 837$). Despite lower volume, SCMIM demonstrated superior engagement quality with highest proportion of on-task messages (90% vs. 85% and 80%) and task completion messages (60% vs. 54% and 50%). ChatGPT-MIM showed highest proportion of interaction messages (35% vs. 28% and 30%), suggesting SCMIM's structured approach enhanced task focus but somewhat constrained spontaneous peer interaction.

These findings indicate that while ChatGPT-MIM promoted higher overall participation rates, SCMIM more effectively maintained task focus and learning-oriented interactions.

Table 3. ANCOVA Results for Behavioral Engagement

Group	Pre-test	Post-test	Adjusted post-test	F	Pairwise comparisons
TG1	3.45 (0.43)	3.89 (0.40)	3.88 (0.08)	10.63***	TG2>TG1*
TG2	3.42 (0.45)	4.12 (0.42)	4.10 (0.08)		TG1>CG***
CG	3.48 (0.41)	3.44 (0.43)	3.43 (0.08)		TG2>CG***

Note. Pre-test and post-test values are Mean (*SD*); adjusted post-test values are Mean (*SE*). *$p < .05$, **$p < .01$, ***$p < .001$.

Table 4. Analysis of Message Patterns

Measure	TG1	TG2	CG	F	Partial η^2
Total messages	952 (124)[b]	1197 (143)[a]	837 (116)[c]	7.85***	.19
Messages per individual	39.7 (6.3)[b]	49.9 (7.5)[a]	34.9 (6.2)[c]	6.92***	.17
Messages per task	158.7 (19.6)[b]	199.5 (25.2)[a]	139.5 (18.8)[c]	8.27***	.19
On-task messages	857 (90%)[a]	1017 (85%)[b]	670 (80%)[c]	8.64***	.20
Off-task messages	95 (10%)[a]	180 (15%)[b]	167 (20%)[c]	8.64***	.2
Task completion messages	514 (60%)[a]	549 (54%)[b]	335 (50%)[c]	5.86**	.15
Interaction messages	240 (28%)[b]	356 (35%)[a]	201 (30%)[a]	5.74**	.14
Others	103 (12%)[a]	112 (13%)[a]	134 (20%)[b]	5.65**	.14

Note. Values are Mean (*SD*) or *n* (%). Different superscripts (a,b,c) indicate significant differences (Tukey's HSD). *p < .05, **p < .01, ***p < .001.

Cognitive Engagement. ANCOVA assumptions were verified (Levene's test: $F(2,69) = 0.79$, $p = .458$; homogeneity of regression slopes: $F(2,66) = 0.38$, $p = .686$). ANCOVA demonstrated significant differences ($F(2,68) = 14.38$, $p < .001$, partial $\eta^2 = .30$), as detailed in Table 5.

SCMIM significantly outperformed both ChatGPT-MIM (mean difference = 0.48, $p = .002$) and traditional MIM (mean difference = 0.86, $p < .001$). ChatGPT-MIM showed significantly higher engagement than traditional MIM (mean difference = 0.38, $p = .004$). Content analysis revealed significant differences in cognitive processing levels (Table 6).

SCMIM showed significantly lower operation-level engagement (28.2%) compared to ChatGPT-MIM (39.2%) and traditional MIM (42.5%). ChatGPT-MIM demonstrated strength in wayfinding (37.5%), exceeding SCMIM (30.4%, $p < .001$).

The most pronounced differences emerged in higher-order activities. SCMIM (27.8%) significantly outperformed both ChatGPT-MIM (16.2%) and traditional MIM (15.4%) in sense-making. For innovation-level thinking, SCMIM (13.6%) showed nearly double the engagement of ChatGPT-MIM (7.1%) and traditional MIM (6.3%). Notably, ChatGPT-MIM and traditional MIM showed no significant differences in higher-order thinking activities (sense-making: $p = .876$; innovation: $p = .792$).

Table 5. ANCOVA Results for Cognitive Engagement

Group	Pre-test	Post-test	Adjusted post-test	*F*	Pairwise comparisons
TG1	3.32 (0.61)	4.25 (0.39)	4.23 (0.08)	14.38***	TG1>TG2**
TG2	3.28 (0.58)	3.76 (0.42)	3.75 (0.08)		TG1>CG***
CG	3.30 (0.60)	3.38 (0.41)	3.37 (0.08)		TG2>CG**

Note. Pre-test and post-test values are Mean (*SD*); adjusted post-test values are Mean (*SE*). *p < .05, **p < .01, ***p < .001.

Table 6. Analysis of Cognitive Processing Levels in Discussions

Level	TG1	TG2	CG	*F*	Partial η^2
Operation	28.2 (4.8)[a]	39.2 (5.1)[b]	42.5 (5.3)[b]	13.67***	.28
Wayfinding	30.4 (5.0)[b]	37.5 (5.2)[a]	35.8 (5.4)[a]	10.84***	.24
Sense-making	27.8 (5.2)[a]	16.2 (4.8)[b]	15.4 (4.5)[b]	12.95***	.27
Innovation	13.6 (3.8)[a]	7.1 (3.4)[b]	6.3 (3.1)[b]	11.87***	.26

Note. Values are Mean (*SD*) of percentage distribution. Different superscripts (a,b) indicate significant differences (Tukey's HSD). *p < .05, **p < .01, ***p < .001.

Emotional Engagement. ANCOVA assumptions were verified (Levene's test: $F(2,69) = 0.86$, $p = .427$; homogeneity of regression slopes: $F(2,66) = 0.40$, $p = .672$). ANCOVA revealed significant differences ($F(2,68) = 13.52$, $p < .001$, partial $\eta^2 = .28$), shown in Table 7.

ChatGPT-MIM demonstrated significantly higher emotional engagement compared to SCMIM (mean difference = 0.34, $p = .004$) and traditional MIM (mean difference = 0.77, $p < .001$). SCMIM also showed significantly higher engagement than traditional MIM (mean difference = 0.43, $p < .001$). Thematic analysis of interviews ($n = 6$ per group) revealed three primary themes (Table 8).

Chi-square analyses revealed significant differences in the distribution of positive experiences across platforms for all three themes, with medium effect sizes (Cramer's V ranging from .33 to .36) as shown in Table 8. Representative student comments highlight these differences:

- Learning Support Experience:
  - SCMIM: *"The structured questions helped me think deeply, but sometimes the constant prompting for deeper reflection felt overwhelming."* (TG1-03)
  - ChatGPT-MIM: *"The instant responses made learning feel more manageable. Whenever I had a question, I knew help was readily available."* (TG2-02)
  - Traditional MIM: *"Often felt lost when trying to understand complex topics without immediate guidance."* (CG-01)
- Discussion Quality:

Table 7. ANCOVA Results for Emotional Engagement

Group	Pre-test	Post-test	Adjusted post-test	*F*	Pairwise comparisons
TG1	3.38 (0.65)	3.82 (0.40)	3.80 (0.07)	13.52***	TG2>TG1***
TG2	3.32 (0.62)	4.15 (0.42)	4.14 (0.07)		TG1>CG***
CG	3.35 (0.64)	3.38 (0.41)	3.37 (0.07)		TG2>CG***

Note. Pre-test and post-test values are Mean (*SD*); adjusted post-test values are Mean (*SE*). *p < .05, **p < .01, ***p < .001.

Table 8. Analysis of Interview Themes by Platform

Theme	TG1 (n = 6)	TG2 (n = 6)	CG (n = 6)	$X^2(2)$	Cramer's V
Learning support experience	67%[b]	83%[a]	33%[c]	10.94**	.34
Discussion quality	75%[a]	83%[a]	42%[b]	11.87**	.33
Learning confidence	58%[b]	92%[a]	33%[c]	12.26**	.36

Note. Values represent percentage of participants expressing positive experiences. Different superscripts (a,b,c) indicate significant differences (Bonferroni-adjusted z-tests). *p < .05, **p < .01, ***p < .001.

  - SCMIM: *"I learned a lot from the structured approach, though it required more emotional energy."* (TG1-06)
  - ChatGPT-MIM: *"The ability to get quick help made me feel more comfortable participating actively."* (TG2-01)
  - Traditional MIM: *"Discussions often became unfocused without proper guidance."* (CG-02)
- Learning Confidence:
  - SCMIM: *"The structured questioning initially felt intimidating but ultimately helped me develop stronger analytical confidence."* (TG1-01)
  - ChatGPT-MIM: *"Always felt supported and comfortable participating, knowing I could get immediate clarification."* (TG2-03)
  - Traditional MIM: *"Often felt uncertain about whether my contributions were on the right track."* (CG-05)

These results demonstrate that SCMIM affects student engagement dimensions differently: while ChatGPT-MIM more effectively promotes behavioral and emotional engagement through immediate support, SCMIM demonstrates superior effectiveness in fostering cognitive engagement, particularly in analytical thinking and knowledge construction.

The Impact on Learners' Perceptions

Analysis of learners' perceptions through the Technology Acceptance Model (TAM) framework revealed distinct patterns across four dimensions:

Perceived Usefulness (PU). Thematic analysis demonstrated significant differences in perceived usefulness across platforms (Table 9).

Both AI-enhanced platforms demonstrated significantly higher perceived usefulness than traditional MIM. SCMIM users particularly valued the structured learning approach: *"The Socratic questioning helped me develop deeper understanding of concepts. The system didn't just give answers but guided my thinking process"* (TG1-02). ChatGPT-MIM users emphasized immediate support benefits: *"Getting instant answers to questions helped maintain learning momentum"* (TG2-02). Traditional MIM

Table 9. Analysis of Perceived Usefulness Themes

Theme	TG1 (n = 6)	TG2 (n = 6)	CG (n = 6)	$X^2(2)$	*Cramer's V*
Learning enhancement	83%[a]	75%[a]	42%[b]	11.24**	.34
Feature contribution	75%[a]	67%[a]	33%[b]	10.65**	.33
Deep learning support	92%[a]	67%[b]	33%[c]	12.84***	.36

Note. Values represent the percentage of participants expressing positive perceptions within each theme. Different superscripts (a,b,c) indicate significant differences between groups based on Bonferroni-adjusted z-tests. *p < .05, **p < .01, ***p < .001.

users noted limitations: *"Basic messaging functions were insufficient for complex learning needs"* (CG-03).

Perceived Ease of Use (PEOU). Analysis revealed notable trade-offs between functionality and complexity (Table 10).

SCMIM users acknowledged adaptation challenges: *"The Socratic questioning framework took time to get used to, but the learning curve was worth it"* (TG1-01). ChatGPT-MIM users consistently reported high ease of use: *"The interface was intuitive, and AI features were easy to access"* (TG2-03). Traditional MIM users found the platform straightforward but functionally limited: *"Very easy to use but lacked advanced learning support features"* (CG-02).

Attitude Towards Use (ATU). Students' attitudes varied significantly based on their learning experiences (Table 11).

SCMIM users reported growing appreciation for the system's pedagogical benefits: *"Initially challenging but grew to value the structured approach for deeper understanding"* (TG1-03). ChatGPT-MIM users showed consistent positive attitudes focused on convenience: *"Consistently helpful for learning and discussion, especially for quick support"* (TG2-04). Traditional MIM users expressed increasingly negative perceptions: *"Initially satisfied but increasingly felt the lack of learning support"* (CG-04).

Behavioral Intention to Use (BIU). Analysis of future use intentions revealed clear preferences for AI-enhanced solutions (Table 12).

Table 10. Analysis of Ease of Use Themes

Theme	TG1 (n = 6)	TG2 (n = 6)	CG (n = 6)	$X^2(2)$	*Cramer's V*
System navigation	58%[b]	92%[a]	92%[a]	12.45***	.36
Learning curve	50%[b]	83%[a]	83%[a]	11.92**	.35
Feature accessibility	67%[b]	83%[a]	92%[a]	10.74**	.33

Note. Values represent the percentage of participants expressing positive perceptions within each theme. Different superscripts (a,b) indicate significant differences between groups based on Bonferroni-adjusted z-tests. *p < .05, **p < .01, ***p < .001.

Table 11. Analysis of Attitude Themes

Theme	TG1 (n = 6)	TG2 (n = 6)	CG (n = 6)	$X^2(2)$	*Cramer's V*
Overall experience	75%[a]	83%[a]	42%[b]	10.86**	.33
Perception evolution	83%[a]	75%[a]	33%[b]	11.64**	.34
Learning value	92%[a]	67%[b]	42%[c]	12.73***	.36

Note. Values represent the percentage of participants expressing positive perceptions within each theme. Different superscripts (a,b,c) indicate significant differences between groups based on Bonferroni-adjusted z-tests. *p < .05, **p < .01, ***p < .001.

SCMIM users emphasized suitability for deep learning contexts: *"Would definitely use this for courses requiring deep understanding and critical thinking"* (TG1-02). ChatGPT-MIM users highlighted versatility: *"Would use this for any course requiring regular discussion and support"* (TG2-02). Traditional MIM users indicated limited future interest: *"Would only use if no better alternatives were available"* (CG-03).

These findings address the second research question, revealing that while both AI-enhanced platforms were perceived more positively than traditional MIM, they exhibited distinct advantages. SCMIM was valued for its contribution to deep learning and critical thinking, despite a steeper learning curve. ChatGPT-MIM was appreciated for its ease of use and immediate support, fostering emotional comfort and participation. The analysis revealed important trade-offs between ease of use and learning depth that should inform system design decisions based on specific educational objectives.

Discussion

The Impact on Engagement

The findings reveal distinctive patterns in how different AI integration approaches influence learner engagement across behavioral, cognitive, and emotional dimensions, offering important insights for designing AI-enhanced educational technologies.

ChatGPT-MIM's superior behavioral engagement (adjusted mean = 4.10 versus 3.88 for SCMIM and 3.43 for traditional MIM) aligns with Ryan and Deci (2020)'s

Table 12. Analysis of Behavioral Intention Themes

Theme	TG1 (n = 6)	TG2 (n = 6)	CG (n = 6)	$X^2(2)$	*Cramer's V*
Future use intent	83%[a]	83%[a]	33%[b]	13.28***	.37
Context suitability	75%[a]	75%[a]	42%[b]	10.93**	.33
Intent to recommend	83%[a]	75%[a]	33%[b]	12.45***	.36

Note. Values represent the percentage of participants expressing positive perceptions within each theme. Different superscripts (a,b) indicate significant differences between groups based on Bonferroni-adjusted z-tests. *p < .05, **p < .01, ***p < .001.

self-determination theory, where immediate AI responses create competence support that reduces perceived participation risk. However, this finding raises important questions about the relationship between participation and learning quality. The higher proportion of off-task messages in ChatGPT-MIM (15% versus 10% in SCMIM) suggests that immediate response availability may inadvertently encourage what Garrison and Cleveland-Innes (2005)'s term "surface interaction", participation that prioritizes activity over meaningful content engagement.

This pattern extends Wu et al. (2023)'s findings on AI integration benefits while illuminating a fundamental tension between accessibility and intellectual rigor. The ease of obtaining immediate responses through ChatGPT-MIM appears to reduce the cognitive friction that often accompanies deep learning, potentially creating a "cognitive bypass effect" where students circumvent the productive struggle that Elder and Paul (2010) identify as essential for knowledge construction. This has important implications for how educators balance immediate support with opportunities for deeper cognitive engagement.

SCMIM's effectiveness in promoting cognitive engagement (adjusted mean = 4.23 versus 3.75 for ChatGPT-MIM and 3.37 for traditional MIM) strongly supports Paul and Elder (2007)'s theoretical claims about systematic questioning's developmental progression. However, our findings extend beyond their traditional classroom applications, demonstrating that these principles can be successfully operationalized through AI technology. The significantly higher rates of sense-making (27.8% versus 16.2% and 15.4%) and innovation-level thinking (13.6% versus 7.1% and 6.3%) suggest that structured AI guidance may offer scalability advantages over traditional approaches, addressing implementation challenges identified in previous Socratic method research while maintaining pedagogical integrity.

Importantly, our finding that ChatGPT-MIM and traditional MIM showed similar distributions in higher-order thinking activities provides crucial evidence about the nature of AI integration effects. While Huang et al. (2025) reported improved student outcomes through basic ChatGPT integration, our cognitive analysis reveals that such improvements may be limited to participation metrics rather than cognitive transformation. This suggests that AI integration without pedagogical structure may represent what Cuban et al. (2001) describes as "technological layering", adding new tools to existing practices without fundamental transformation of learning processes. This insight has important implications for how institutions approach AI integration in educational contexts.

The Socratic framework implemented in SCMIM functions as what Vygotsky (1978) conceptualized as a mediating tool within the zone of proximal development, but our implementation demonstrates possibilities beyond traditional human-mediated scaffolding. By structuring responses to include assumption examination and implication exploration, SCMIM creates cognitive scaffolding that operates consistently and systematically, addressing Wu et al. (2025)'s concern about the unpredictability of AI responses in educational settings. This demonstrates how pedagogical frameworks can transform AI from a simple information source into a tool for systematic cognitive development.

ChatGPT-MIM's higher emotional engagement scores (adjusted mean = 4.14 versus 3.80 for SCMIM) align with Lee et al. (2023)'s findings about AI's emotionally supportive functions, yet our results reveal important temporal complexities absent from previous research. While immediate support creates an "emotionally supportive scaffold," our interview data revealed that SCMIM users developed growing appreciation for structured challenge over time, a finding that challenges common assumptions about student preference for ease over difficulty. This supports Steenhof et al. (2020)'s concept of "productive struggle" but demonstrates its applicability in AI-mediated environments, suggesting that students can recognize and value intellectual challenge when it serves clear learning purposes.

These multidimensional engagement patterns reveal a fundamental trade-off that Faranda et al. (2020) distinguish as surface versus deep approaches to learning, but our findings suggest this trade-off may be more nuanced than previously understood. While ChatGPT-MIM facilitates broader participation and emotional comfort, SCMIM more effectively promotes what Entwistle (2017)'s terms "deep learning strategies." Rather than viewing these as competing outcomes, our findings suggest that different AI approaches may serve complementary functions in supporting comprehensive learning experiences, with important implications for how educators sequence and deploy different AI integration approaches across learning contexts.

The Impact on Student Perceptions

The divergent perception patterns between SCMIM and ChatGPT-MIM reveal sophisticated student awareness of different pedagogical affordances, offering important insights for understanding educational technology adoption.

The pronounced difference in perceived usefulness for deep learning support (92% for SCMIM versus 67% for ChatGPT-MIM) challenges common assumptions in the educational technology literature that students primarily value convenience and immediate gratification. This finding extends Lee et al. (2024)'s work on student preferences, instead demonstrating student capacity to recognize and value more sophisticated pedagogical approaches when they experience clear learning benefits.

Our finding that SCMIM achieved comparable behavioral intention to use scores (83%) as ChatGPT-MIM despite significantly lower perceived ease of use (58% versus 92%) offers important insights for technology acceptance theory in educational contexts. This pattern differs from traditional TAM predictions and most previous educational technology research, which have consistently documented the importance of ease-of-use preferences (Chen & Aklikokou, 2020). Our results suggest that Davis (1989)'s original framework, developed for workplace efficiency technologies, may need refinement for educational contexts where learning transformation represents the primary value proposition. This has implications for how we conceptualize and measure technology acceptance in educational settings.

The temporal evolution of attitudes toward SCMIM, from initial resistance to growing appreciation, presents an important consideration for educational technology implementation. While Huang et al. (2025) reported sustained positive attitudes toward straightforward ChatGPT implementations, our findings suggest that pedagogically sophisticated systems may require extended evaluation periods to reveal their full acceptance potential. This challenges the rapid assessment methods common in educational technology research and suggests that immediate user satisfaction may not fully capture the value of tools designed for longer-term learning development.

These perception patterns align with emerging research by Chen and Aklikokou (2020) showing that for technologies with clear developmental benefits, perceived learning value can override ease of use considerations. However, our findings extend this work by demonstrating that such value recognition may develop over time rather than being immediately apparent. This has important implications for how institutions introduce and support adoption of pedagogically sophisticated educational technologies, suggesting the need for adequate orientation, ongoing support, and patience during initial adaptation periods.

The Complementary Value of Different AI Integration Approaches

Rather than viewing these systems as competing alternatives, our findings suggest a more nuanced understanding of how different AI approaches serve fundamentally different pedagogical functions.

ChatGPT-MIM's effectiveness for "engagement scaffolding" extends Annamalai (2019)'s confidence-building concepts to AI-mediated environments. However, our findings reveal important considerations about the boundaries of this approach. While immediate AI support reduces participation barriers, the cognitive engagement limitations suggest that this approach may create what we term "pedagogical ceiling effects", a point beyond which additional AI support does not enhance learning depth. This has important implications for understanding when and how to deploy immediate-support AI tools in educational contexts.

SCMIM's function as "cognitive elevation technology" supports Elder and Paul (2010)''s theoretical framework but demonstrates practical implementation possibilities that extend previous research. The systematic cognitive improvements achieved through structured AI guidance suggest scalability potential that human-only Socratic implementations cannot match, addressing Lodge et al. (2018)'s concerns about resource limitations in implementing sophisticated pedagogical approaches. This demonstrates how AI technology can make established pedagogical approaches more accessible and sustainable across diverse educational contexts.

The complementary nature of these strengths challenges the competitive framing common in educational technology literature, instead supporting context-sensitive deployment strategies that leverage different AI approaches for different learning needs. This finding offers important guidance for educational designers considering how to sequence or combine different AI integration approaches across courses, units,

or learning phases. For example, ChatGPT-MIM might effectively establish participation confidence and emotional safety early in a course, while SCMIM could be introduced progressively as discussions advance to more complex analytical topics. Such strategic deployment recognizes that different learning objectives and student readiness levels may benefit from different forms of AI support.

Implications for Theory and Practice

These findings offer important refinements to existing frameworks about AI integration in educational contexts while providing practical guidance for implementation.

Theoretically, our demonstration that pedagogical structure in AI interactions systematically elevates learning depth extends current understanding of AI's role in education. While recent studies (Huang et al., 2025; Wu et al., 2023) have focused primarily on AI's technological capabilities, our findings suggest that educational value depends critically on pedagogical intentionality rather than technological sophistication alone. This insight has important implications for how we conceptualize and design AI-enhanced educational technologies, suggesting that pedagogical frameworks should guide technological implementation rather than the reverse.

The similar higher-order thinking distributions between ChatGPT-MIM and traditional MIM provide important evidence regarding assumptions about AI's inherent educational benefits. This finding suggests that AI integration alone may primarily amplify existing educational practices rather than transform them, supporting more nuanced perspectives about technology's role in education. For practitioners and policymakers, this underscores the importance of focusing on how AI tools are pedagogically structured rather than simply whether AI is integrated.

For practitioners, our findings offer guidance for resolving apparent tensions in previous research about student preferences for challenge versus support. The evolution of student perceptions toward SCMIM suggests that resistance to sophisticated pedagogical approaches may be temporary rather than fixed, developing into appreciation as students experience learning benefits over time. This has important implications for implementation strategies, suggesting that institutions should provide adequate orientation and support during initial adaptation periods rather than abandoning sophisticated approaches based on early user resistance.

The systematic integration of Socratic principles within SCMIM demonstrates how traditional pedagogical frameworks can be operationalized through contemporary technologies, addressing the implementation challenges identified in previous Socratic method research while maintaining pedagogical integrity. This integration model provides empirical evidence for theoretical claims about technology-pedagogy integration that have previously lacked robust experimental support. For educational technology developers, this offers a concrete example of how established pedagogical principles can be translated into AI system design

through careful prompt engineering and response structuring. For educators, it demonstrates that sophisticated pedagogical approaches traditionally requiring intensive instructor involvement can be scaled through thoughtful AI integration, making them more accessible across diverse educational contexts and resource constraints.

Conclusion

This study investigated the impact of Socratic-based ChatGPT integration (SCMIM), standard ChatGPT integration (ChatGPT-MIM), and traditional mobile instant messaging (MIM) on student engagement and perceptions in blended learning discussions. Through a 12-week randomized controlled trial with 72 master's students, we examined how different AI integration approaches influence behavioral, cognitive, and emotional engagement alongside perceptions based on the Technology Acceptance Model (TAM).

Key Contributions

This study provides three principal contributions that fundamentally challenge prevailing assumptions about AI integration in educational contexts. First, we demonstrated that pedagogical frameworks, rather than technological sophistication, fundamentally determine educational outcomes. The integration of Socratic methodology transformed identical AI technology into a more effective scaffold for higher-order thinking, with significant increases in sense-making activities (27.8% versus 16.2% for ChatGPT-MIM) and innovation-level thinking (13.6% versus 7.1%). This finding contradicts techno-optimistic assumptions prevalent in current literature that emphasize technological capabilities over pedagogical intentionality.

Second, we identified complementary rather than competing strengths across different AI integration approaches, challenging the either-or framing common in educational technology research. ChatGPT-MIM excelled in promoting behavioral participation (adjusted mean = 4.10 versus 3.88 for SCMIM) and emotional comfort (adjusted mean = 4.14 versus 3.80), while SCMIM more effectively developed advanced cognitive processes (adjusted mean = 4.23 versus 3.75). Crucially, our finding that ChatGPT-MIM and traditional MIM showed similar distributions in higher-order thinking activities reveals that basic AI integration may primarily amplify existing educational practices rather than transform them, a critical insight absent from previous literature.

Third, we revealed important contradictions to established technology acceptance theory through student perceptions of pedagogically-structured AI tools. Despite requiring greater initial adaptation, SCMIM achieved comparable behavioral intention to use scores as ChatGPT-MIM (both 83%), demonstrating that students valued its superior contribution to deep learning (92% versus 67%) even when it required additional cognitive investment. This finding directly challenges Davis's (1989) Technology Acceptance Model assumptions about the primacy of ease of use, suggesting that

educational technology adoption may follow fundamentally different patterns than workplace technology adoption.

Limitations

This study encountered several specific limitations that directly impact the generalizability and interpretation of our findings. First, our Socratic implementation within SCMIM was constrained by the single-response architecture of mobile instant messaging platforms. Unlike human-led Socratic dialogue that adapts dynamically based on student responses, our system relied on predetermined prompt structures that embedded Socratic principles within comprehensive single responses. While this approach successfully promoted cognitive engagement, it could not replicate the iterative questioning sequences that characterize expert Socratic facilitation, potentially limiting the depth of cognitive provocation achievable through the system.

Second, our measurement approach introduced specific methodological constraints that may have influenced our findings. The cognitive engagement analysis relied on content coding of observable discussion posts rather than direct assessment of internal cognitive processes, meaning that students who engaged in sophisticated thinking without articulating it in discussions remained undetected by our measures. Additionally, our reliance on self-reported engagement questionnaires may have been influenced by social desirability bias, particularly given that students were aware of their group assignments and the experimental nature of the study.

Third, the 12-week implementation period, while sufficient to detect significant differences, may have been insufficient to fully capture the long-term adaptation processes evident in student perceptions. Our interview data revealed evolving attitudes toward SCMIM's structured approach, suggesting that longer implementation periods might yield different engagement and perception patterns. Furthermore, the timing of our post-intervention measurements (immediately following the 12-week period) may not have captured the full extent of students' adaptation to different AI integration approaches.

Fourth, our participant population, 72 master's students enrolled in an Educational Technology Applications course at a Taiwanese university, represents a specific demographic with particular characteristics that limit generalizability. These students possessed prior experience with educational technology and were predisposed toward critical analysis of technological tools, potentially making them more receptive to SCMIM's structured approach than students in other disciplines or educational levels. Additionally, the course content naturally emphasized critical thinking and technology evaluation, creating an optimal environment for Socratic methodology that may not exist in other educational contexts.

Fifth, our exclusive use of the Line messaging platform introduces platform-specific constraints that may not apply to other messaging environments. Line's interface characteristics, user familiarity among Taiwanese students, and integration capabilities with our AI systems may have influenced engagement patterns in ways that would not

replicate across different messaging platforms or cultural contexts. The generalizability of our findings to other mobile instant messaging platforms therefore remains uncertain.

Finally, our investigation of student perceptions focused exclusively on core TAM dimensions without incorporating contextual factors that may significantly influence educational technology acceptance. Variables such as instructor support, peer influence, course requirements, and institutional technology policies were not systematically examined, despite their potential impact on student attitudes and usage intentions. This limitation is particularly significant given our finding that traditional TAM predictions were not fully supported in this educational context.

Future Research

Several critical research directions emerge from this study's findings and limitations. First, future research should investigate adaptive Socratic implementations that can adjust questioning depth and complexity based on real-time analysis of student responses. Such systems could potentially bridge the gap between predetermined prompt structures and dynamic human-led facilitation while maintaining the scalability advantages of AI-mediated instruction.

Second, longitudinal studies examining the temporal evolution of engagement and perceptions across extended implementation periods would provide crucial insights into the adaptation processes we observed. Research tracking student responses over full academic semesters or years could reveal whether the initial resistance to structured pedagogical approaches consistently transforms into appreciation, or whether individual differences predict varied adaptation trajectories.

Third, while this study demonstrated significant differences in cognitive engagement patterns, future research should directly assess actual learning outcomes through performance metrics such as examination scores, assignment quality, or knowledge retention tests. Incorporating follow-up assessments that measure long-term knowledge application and transfer would strengthen claims about deeper learning effects. Such studies could employ pre-post achievement tests, concept mapping exercises, or authentic performance tasks to triangulate engagement indicators with concrete learning gains, providing more comprehensive evidence of pedagogical effectiveness.

Fourth, comparative investigations across different disciplinary contexts are essential for understanding the boundary conditions of Socratic-based AI integration effectiveness. Disciplines with different epistemological traditions, such as mathematics, natural sciences, or creative arts, may require modified approaches to structured questioning that align with their specific ways of knowing and argumentation patterns.

Fifth, research should explore hybrid implementation strategies that leverage the complementary strengths identified in this study. Systems that begin with ChatGPT-MIM's approach to establish participation confidence and emotional safety, then progressively introduce SCMIM's structured elements as discussions

advance, could potentially optimize both engagement and cognitive development outcomes.

Finally, studies employing larger and more diverse participant populations across multiple institutions, educational levels, and cultural contexts would significantly enhance generalizability of findings. Cross-cultural investigations examining how different educational traditions and cultural contexts influence receptiveness to AI-mediated Socratic questioning are particularly critical for global implementation. The effectiveness of structured questioning approaches may vary significantly across cultural contexts with different pedagogical traditions and student-teacher relationship expectations. Expanding research to include undergraduate students, different academic disciplines, and varied institutional settings would provide robust evidence for broader applicability of Socratic-enhanced AI integration approaches.

Concluding remarks

This study demonstrates that effective AI integration in educational discussions requires fundamental reconsideration of the relationship between technological capabilities and pedagogical intentions. Rather than assuming that more sophisticated AI technology automatically produces better educational outcomes, our findings reveal that pedagogical structure, specifically the systematic incorporation of Socratic principles, transforms AI from a simple information source into a cognitive scaffolding tool that promotes deeper learning outcomes.

The implications extend beyond technological implementation to challenge core assumptions about student preferences and educational technology design. Our finding that students valued pedagogically sophisticated AI approaches despite their greater complexity suggests that educational technology research has potentially underestimated student capacity for recognizing and appreciating intellectual challenge when it serves clear learning purposes.

As educational institutions increasingly adopt blended learning approaches and AI technologies, understanding how pedagogical frameworks can enhance rather than compete with technological affordances becomes crucial for realizing the transformative potential of AI in education. This study provides empirical evidence that such integration is not only possible but necessary for developing educational technologies that support comprehensive intellectual development rather than merely efficient information access.

The future of AI in education lies not in maximizing technological sophistication but in thoughtful integration of proven pedagogical principles with technological capabilities, an integration that respects both the power of structured educational guidance and the scalability potential of AI systems. This balanced approach offers a promising path toward educational technologies that truly enhance rather than simply digitize traditional learning processes.

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ORCID iDs

Hsin-Yu Lee https://orcid.org/0000-0003-3257-305X
Pei-Hua Chen https://orcid.org/0009-0002-0431-4499
Yueh-Min Huang https://orcid.org/0000-0001-7052-1272
Ting-Ting Wu https://orcid.org/0000-0003-4970-7042

Authors' Contributions

Hsin-Yu Lee is the leader of this research, he is in charge of the research design, conducting teaching and learning experiment, data analysis.
Pei-Hua Chen is responsible for assisting in the conduct of experiments and surveying related literature, writing the manuscript, and proofreading the manuscript.
Yueh-Min Huang is responsible for designing research experiments, providing fundamental education theories and comments to this research, and he is also responsible for revising the manuscript. All authors spent more than 2 months to discuss and analyze the data. The author(s) read and approved the final manuscript.
Ting-Ting Wu is responsible for assisting in the conduct of experiments and surveying related literature.

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Data Availability Statement

The datasets utilized in this research can be obtained from the corresponding author upon reasonable request.

References

Allison, J. (2025). RAISE the standard: A framework for transparent reporting of artificial intelligence studies in education. *Journal of Educational Computing Research*, *64*(1). https://doi.org/10.1177/07356331251377430

Annamalai, N. (2019). Using WhatsApp to extend learning in a blended classroom environment. *Teaching English with Technology*, *19*(1), 3–20.

Ardèvol-Abreu, A., Patricia, D., Helena, B., & Rodríguez-Wangüemert, C. (2023). Mobile instant messaging uses and technostress: A qualitative approach. *International Journal of Human-Computer Interaction*, *39*(15), 3003–3015. https://doi.org/10.1080/10447318.2022.2090525

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Chen, L., & Aklikokou, A. K. (2020). Determinants of E-government adoption: Testing the mediating effects of perceived usefulness and perceived ease of use. *International Journal of Public Administration*, *43*(10), 850–865. https://doi.org/10.1080/01900692.2019.1660989

Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*. Routledge.

Cuban, L., Kirkpatrick, H., & Peck, C. (2001). High access and low use of technologies in high School classrooms: Explaining an apparent paradox. *American Educational Research Journal*, *38*(4), 813–834. https://doi.org/10.3102/00028312038004813

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, *13*(3), 319–340. https://doi.org/10.2307/249008

Elder, L., & Paul, R. (2010). Critical thinking: Competency standards essential for the cultivation of intellectual skills, Part 1. *Journal of Developmental Education*, *34*(2), 38–39. https://www.proquest.com/scholarly-journals/critical-thinking-competency-standards-essential/docview/1285469423/se-2

Entwistle, N. (2017). *Teaching for understanding at university: Deep approaches and distinctive ways of thinking*. Bloomsbury Publishing.

Etori, N. A., & Gini, M. (2024). WisCompanion: Integrating the Socratic Method with ChatGPT-Based AI for Enhanced Explainability in Emotional Support for Older Adults. In: Degen, H., Ntoa, S. (eds) *Artificial Intelligence in HCI*. HCII 2024. Lecture Notes in Computer Science (vol 14734). Springer, Cham. https://doi.org/10.1007/978-3-031-60606-9_11

Faranda, W. T., Clarke, T. B., & ClarkeIii, I. (2020). Marketing student perceptions of academic program quality and relationships to surface, deep, and strategic learning approaches. *Journal of Marketing Education*, *43*(1), 9–24. https://doi.org/10.1177/0273475320939261

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, *39*(2), 175–191. https://doi.org/10.3758/BF03193146

Fisher, R., Perényi, Á., & Birdthistle, N. (2018). The positive relationship between flipped and blended learning and student engagement, performance and satisfaction. *Active Learning in Higher Education*, *22*(2), 97–113. https://doi.org/10.1177/1469787418801702

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, *74*(1), 59–109. https://doi.org/10.3102/00346543074001059

Garrison, D. R., & Cleveland-Innes, M. (2005). Facilitating cognitive presence in online learning: Interaction is not enough. *American Journal of Distance Education*, *19*(3), 133–148. https://doi.org/10.1207/s15389286ajde1903_2

Huang, Y.-M., Chen, P.-H., Lee, H.-Y., Sandnes, F. E., & Wu, T.-T. (2025). ChatGPT-enhanced mobile instant messaging in online learning: Effects on student outcomes and perceptions. *Computers in Human Behavior*, *168*, Article 108659. https://doi.org/10.1016/j.chb.2025.108659

Huberty, C. J., & Morris, J. D. (1992). Multivariate analysis versus multiple univariate analyses. In *Methodological issues & strategies in clinical research* (pp. 351–365). American Psychological Association. https://doi.org/10.1037/10109-030

Labadze, L., Grigolia, M., & Machaidze, L. (2023). Role of AI chatbots in education: Systematic literature review. *International Journal of Educational Technology in Higher Education*, *20*(1), 56. https://doi.org/10.1186/s41239-023-00426-1

Lee, H.-Y., Chen, P.-H., Wang, W.-S., Huang, Y.-M., & Wu, T.-T. (2024). Empowering ChatGPT with guidance mechanism in blended learning: Effect of self-regulated learning, higher-order thinking skills, and knowledge construction. *International Journal of Educational Technology in Higher Education*, *21*(1), 16. https://doi.org/10.1186/s41239-024-00447-4

Lee, H.-Y., & Huang, Y.-M. (2024). Bridging STEM Education and Ubiquitous Learning: A Case Study on Developing a LINE Chatbot with Google's Gemini for Virtual Peer Collaboration. In: Cheng, YP., Pedaste, M., Bardone, E., Huang, YM. (eds) *Innovative Technologies and Learning*. ICITL 2024. Lecture Notes in Computer Science, vol 14786. Springer, Cham. https://doi.org/10.1007/978-3-031-65884-6_25

Lee, H.-Y., Huang, Y.-M., & Wu, T.-T. (2025). ChatGPT in education: A systematic review of current landscape, limitations and future directions through general system theory lens. *European Journal of Education*, *60*(4), Article e70262. https://doi.org/10.1111/ejed.70262

Lee, H.-Y., Lin, C.-J., Wang, W.-S., Chang, W.-C., & Huang, Y.-M. (2023). Precision education via timely intervention in K-12 computer programming course to enhance programming skill and affective-domain learning objectives. *International Journal of STEM Education*, *10*(1), 52. https://doi.org/10.1186/s40594-023-00444-5

Lee, H.-Y., & Wu, T.-T. (2025). Enhancing blended learning discussions with a Scaffolded Knowledge Integration–Based ChatGPT mobile instant messaging system. *Computers & Education*, *237*, Article 105375. https://doi.org/10.1016/j.compedu.2025.105375

Li, J., Zhang, S., & Ao, W. (2023). Why is instant messaging not instant? Understanding users' negative use behavior of instant messaging software. *Computers in Human Behavior*, *142*, Article 107655. https://doi.org/10.1016/j.chb.2023.107655

Lodge, J., Kennedy, G., Lockyer, L., Arguel, A., & Pachman, M. (2018). Understanding difficulties and resulting confusion in learning: An integrative review. *Frontiers in Education*, *3*, 49. https://doi.org/10.3389/feduc.2018.00049

Mali, D., & Lim, H. (2021). How do students perceive face-to-face/blended learning as a result of the Covid-19 pandemic? *International Journal of Management in Education*, *19*(3), Article 100552. https://doi.org/10.1016/j.ijme.2021.100552

Oyler, D. R., & Romanelli, F. (2014). The fact of ignorance revisiting the socratic method as a tool for teaching critical thinking. *American Journal of Pharmaceutical Education*, *78*(7), 144. https://doi.org/10.5688/ajpe787144

Patel, N. S. (2022). Development of criticality in thought: A conceptual framework for online student discussion forums in higher education. https://doi.org/10.46451/ijts.2021.09.02

Paul, R., & Elder, L. (2007). Critical thinking: The art of socratic questioning. *Journal of Developmental Education*, *31*(1), 36–37.

Ryan, R. M., & Deci, E. L. (2020). Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemporary Educational Psychology*, *61*, Article 101860. https://doi.org/10.1016/j.cedpsych.2020.101860

Shang, H.-F. (2021). The impact of blended discussions on learning motivation and EFL reading performance. *Educational Studies*, *57*(6), 629–649. https://doi.org/10.1080/00131946.2021.1969932

Steenhof, N., Woods, N. N., & Mylopoulos, M. (2020). Exploring why we learn from productive failure: Insights from the cognitive and learning sciences. *Advances in Health Sciences Education*, *25*(5), 1099–1106. https://doi.org/10.1007/s10459-020-10013-y

Steinberg, M. (2020). LINE as super app: Platformization in East Asia. *Social Media + Society*, *6*(2), Article 2056305120933285. https://doi.org/10.1177/2056305120933285

Sun, J. C.-Y., & Rueda, R. (2012). Situational interest, computer self-efficacy and self-regulation: Their impact on student engagement in distance education. *British Journal of Educational Technology*, *43*(2), 191–204. https://doi.org/10.1111/j.1467-8535.2010.01157.x

Tang, Y., & Hew, K. F. (2017). Is mobile instant messaging (MIM) useful in education? Examining its technological, pedagogical, and social affordances. *Educational Research Review*, *21*, 85–104. https://doi.org/10.1016/j.edurev.2017.05.001

Tang, Y., & Hew, K. F. (2022). Effects of using mobile instant messaging on student behavioral, emotional, and cognitive engagement: A quasi-experimental study. *International Journal of Educational Technology in Higher Education*, *19*(1), 3. https://doi.org/10.1186/s41239-021-00306-6

Tomei, L. A., Maine, J., Moussa, K., Holler, M. B., Hobbs, B., & Austin, S. (2024). The top 12 technologies for teaching and learning in the post-pandemic era. In *Exploring technology-infused education in the post-pandemic era* (pp. 1–95). IGI Global. https://doi.org/10.4018/979-8-3693-2885-9.ch001

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (*86*). Harvard University Press.

Wu, T.-T., Lee, H.-Y., Chen, P.-H., Lin, C.-J., & Huang, Y.-M. (2025). Integrating peer assessment cycle into ChatGPT for STEM education: A randomised controlled trial on knowledge, skills, and attitudes enhancement. *Journal of Computer Assisted Learning*, *41*(1), Article e13085. https://doi.org/10.1111/jcal.13085

Wu, T.-T., Lee, H.-Y., Li, P.-H., Huang, C.-N., & Huang, Y.-M. (2023). Promoting self-regulation progress and knowledge construction in blended learning via ChatGPT-based learning aid. *Journal of Educational Computing Research*, *61*(8), 3–31. https://doi.org/10.1177/07356331231191125

Yen, S.-C., Lo, Y., Lee, A., & Enriquez, J. (2018). Learning online, offline, and in-between: Comparing student academic outcomes and course satisfaction in face-to-face, online, and

blended teaching modalities. *Education and Information Technologies*, *23*(5), 2141–2153. https://doi.org/10.1007/s10639-018-9707-5

Zimba, Z. F., Khosa, P., & Pillay, R. (2021). Using blended learning in South African social work education to facilitate student engagement. *Social Work Education*, *40*(2), 263–278. https://doi.org/10.1080/02615479.2020.1746261

Author Biographies

Hsin-Yu Lee received his B.S. degree from the Department of Technology Application and Human Resource Development at National Taiwan Normal University, Taiwan, R.O.C., in 2019. He then obtained his M.S. degree from the Department of Engineering Science at National Cheng Kung University, Taiwan, R.O.C., in 2021. In 2024, he completed his Ph.D. in Engineering Science at National Cheng Kung University, Taiwan, R.O.C. Currently, Dr. Lee is a postdoctoral researcher at the Graduate School of Technological and Vocational Education, National Yunlin University of Science and Technology, Taiwan, R.O.C. His research interests include educational technology, learning analytics, computer vision, and artificial intelligence.

Pei-Hua Chen received her B.S. degrees from the Department of Cultural and Creative Industries, National Chin-Yi University of Technology, Taiwan, R.O.C., in 2023. She is currently pursuing her PhD degree in engineering science at National Cheng Kung University, Taiwan. Her research interests include educational technology, design thinking, UI/UX design and artificial intelligence.

Yueh-Min Huang received the M.S. and Ph.D. degrees in Electrical Engineering from the University of Arizona in 1988 and 1991 respectively. He is a Chair Professor in Department of Engineering Science, National Cheng-Kung University, Taiwan. He has co-authored 3 books and has published more than 250 refereed journal research papers. His research interests include e-Learning, multimedia communications, wireless networks, and artificial intelligence. He is in the editorial board of several international journals in the area of educational technology, computer communications, and web intelligence, including 3 SSCI-indexed e-Learning ones. In e-Learning area, except of publishing more than 100 SSCI-indexed journal papers and editing 3 special issues in SSCI-indexed journals, he was also serving as the directors of Disciplines of Applied Science Education and Innovative Engineering Education in Taiwan's Ministry of Science and Technology. Dr. Huang is a senior member of the IEEE and became Fellow of British Computer Society in 2011. Dr. Huang has received many research awards, such as Taiwan's National Outstanding Research Award in 2011 and 2014, given to Taiwan's top 100 scholars. According to a paper published in BJET, he is ranked no. 3 in the world on terms of the number of educational technology papers published in the period 2012 to 2017.

Ting-Ting Wu is a distinguished professor and director of the Graduate School of Technological and Vocational Education, National Yunlin University of Science and Technology, Taiwan. Her research covers such areas as technology-enhanced language learning and applications of advanced technology to support nursing education.

Appendix

Appendix A.

RAISE 2025 Ethics and Risk Matrix

Risk theme	Risk category	Description	Observed or anticipated impact (author to specify)	Mitigation/safeguards (author to specify)
Pedagogical integrity and learning impact	Loss of learner agency	Potential over-reliance on AI that may reduce learner autonomy and critical thinking.	SCMIM's structured questioning initially reduced some students' sense of autonomy, with learners feeling "guided" rather than self-directing their inquiry	Implemented progressive scaffolding allowing increased autonomy as students adapted; provided optional AI function usage; monitored student autonomy through interviews
	Misuse or overuse of AI	Risk of AI being used beyond intended scope or without adequate oversight.	ChatGPT-MIM group showed higher AI dependency, potentially over-relying on immediate answers rather than independent thinking	Designed SCMIM using socratic questioning rather than direct answers; encouraged peer interaction to complement AI support; limited AI use to specific functions (@reflect, @synthesize)
	Emotional impact and wellbeing	Potential negative emotional effects from AI feedback (e.g., anxiety, discouragement).	Initial adaptation challenges with SCMIM created cognitive load and pressure (58% reported difficulty during early stages based on ease of use perceptions)	Provided 3-week comprehensive training period; continuously monitored emotional responses through interviews; designed progressive difficulty in questioning
	Data misinterpretation	Risk that educators or learners may misinterpret AI-generated analytics or feedback.	Students might misinterpret the deeper meaning of AI-generated socratic reflection prompts	Provided clear explanations within system responses; supplemented with face-to-face discussions; instructors regularly reviewed AI content for pedagogical appropriateness
	Hallucination/ Misinformation	Risk that AI may generate incorrect or misleading information.	AI could generate inaccurate educational content or misleading question frameworks	Used GPT-4o-mini with carefully calibrated temperature settings (0.3-0.4) to enhance accuracy; instructors regularly audited AI response quality; encouraged students to cross-verify information
	Transparency and explainability	Lack of clarity about how AI feedback or decisions are generated.	Students initially unclear about how AI generates structured socratic responses	Explained system operation principles before study commencement; provided system design documentation; clarified AI questioning logic during training phase

(continued)

Appendix A. (continued)

Risk theme	Risk category	Description	Observed or anticipated impact (author to specify)	Mitigation/safeguards (author to specify)
Equity, inclusion, and accessibility	Bias and fairness	Risk that AI may perpetuate or amplify biases (e.g., socio-economic, gender, race).	AI responses might inadvertently reflect biases from training data, potentially affecting students from different backgrounds	Random assignment ensured balanced groups; monitored participation patterns across gender; examined potential bias through content analysis
	Cultural insensitivity	AI outputs or prompts that may be culturally inappropriate or irrelevant.	English AI responses and western educational frameworks may not align with taiwanese cultural context	Conducted research within taiwanese educational context considering local cultural adaptation; used familiar line platform; instructors could intervene for culturally inappropriate content
	Accessibility barriers	Risks of excluding learners with disabilities or diverse needs.	System reliance on smartphones and internet connectivity might exclude some learners	Selected widely-used line platform among students; verified all participants had necessary devices; provided technical support
	Equity in access to technology	Unequal access to devices or connectivity impacting participation.	While master's students generally have devices, varying proficiency levels might affect learning outcomes	Baseline data verified no significant differences in technology proficiency across groups; provided 3-week technical training; ensured equal technical support for all groups
Data governance and learner rights	Privacy and data security	Risks related to unauthorized access, data leakage, or privacy breaches.	Discussion content and personal learning data stored through line and firebase platforms	Obtained university research ethics committee approval; participants signed informed consent; data anonymization; secure firebase storage
	Informed consent and autonomy	Concerns over learner awareness of AI involvement and data use.	Students needed clear understanding of AI's role and data usage in learning process	Provided comprehensive research explanation before study; obtained written informed consent; allowed withdrawal at any time; clarified data usage scope
	Data ownership and usage rights	Uncertainties about who owns student-generated data and AI outputs.	Ownership of student discussion content and AI interaction data unclear	Informed consent explicitly stated data ownership and usage purposes; committed to research-only use; anonymized for publication; students could request data deletion

(continued)

Appendix A. (continued)

Risk theme	Risk category	Description	Observed or anticipated impact (author to specify)	Mitigation/safeguards (author to specify)
Technical dependability and system reliability	Technical failures/ Downtime	Possible interruptions due to AI system malfunctions or outages.	Line platform, ChatGPT API, or firebase could experience failures affecting learning continuity	Used stable commercial platforms (line, OpenAI); established technical support team; prepared backup communication plans; regular system monitoring
	Dependence on proprietary systems	Concerns about transparency, replicability, and vendor lock-in with closed-source AI.	Research depends on OpenAI's GPT-4o-mini and line platform proprietary technologies	Documented system design and prompt engineering methods in detail to facilitate replication; provided technical implementation details; discussed alternative platform possibilities
	Algorithmic accountability	Unclear assignment of responsibility for AI-generated outcomes or errors.	Unclear responsibility attribution when AI generates inappropriate or incorrect responses	Instructors continuously supervised AI-generated content; established problem reporting mechanism; research team accepted responsibility for AI outputs in study design; retained human intervention authority